

2010376103 CSE4261 Assignment 1

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1 Dataset Description

The CIFAR-100 dataset contains 60,000 color images of size 32×32 , grouped into 100 classes with 600 images per class. In this study, only the first 20 classes (class indices 0–19) were used to reduce computational cost. The data were split into training and test sets as provided by the dataset. Pixel values were normalized to the $[0,1]$ range, and labels were converted into one-hot vectors with 20 output categories.

2 Experimental Setup

Ten convolutional neural network (CNN) architectures pre-trained on the ImageNet dataset were evaluated. The models included Xception, VGG16, VGG19, ResNet50, ResNet50V2, ResNet101, ResNet101V2, ResNet152, ResNet152V2, and MobileNet. For each model, the convolutional base was loaded without its top classification layers (`include_top=False`). A custom classifier was attached to adapt the networks for 20-class classification. The classifier consisted of:

- Flatten layer
- Dropout (0.4)
- Dense(512, ReLU) + BatchNormalization
- Dense(256, ReLU) + BatchNormalization
- Dropout(0.4)
- Dense(20, Softmax)

All models used ImageNet weights for initialization. The convolutional base was kept frozen during training, while only the new dense layers were trained on the 20-class CIFAR-100 subset. Training employed the Adam optimizer, categorical cross-entropy loss, and early stopping based on validation performance. Model performance was compared in terms of accuracy, parameter size, and inference time on CPU and GPU.

3 Model Comparison

Table 1 presents the performance of the ten pre-trained CNNs. MobileNet achieved the highest accuracy (64.40%) with the smallest size (14.85 MB) and lowest inference time on GPU. Xception closely followed (64.25%) but was significantly larger. VGG16 and VGG19 showed poor generalization despite moderate size. ResNet50 and ResNet101 offered balanced performance (around 57% accuracy) with higher parameter counts. The V2 variants underperformed, suggesting less transferability of their learned features to low-resolution data.

Table 1: Comparison of pre-trained CNNs on 20 CIFAR-100 classes

No	Architecture	Accuracy (%)	Hardware	Params & size (MB)
1	Xception	64.25	CPU	22,050,108 (84.11)
2	VGG16	17.00	CPU	15,116,884 (57.67)
3	VGG19	17.19	GPU	20,426,580 (77.92)
4	ResNet50	57.25	CPU	24,776,340 (94.51)
5	ResNet50V2	18.75	CPU	24,753,428 (94.43)
6	ResNet101	57.80	CPU	43,846,804 (167.26)
7	ResNet101V2	12.03	GPU	43,815,188 (167.14)
8	ResNet152	47.72	GPU	59,559,572 (227.20)
9	ResNet152V2	7.23	GPU	59,520,276 (227.05)
10	MobileNet	64.40	GPU	3,893,204 (14.85)

4 Summary

Among the compared CNNs, MobileNet demonstrates the best trade-off between accuracy, computational efficiency, and model size. Xception offers comparable accuracy but at higher cost. ResNet architectures maintain moderate accuracy with larger storage requirements, while VGG and V2 variants perform poorly on small, low-resolution images, indicating limited transfer learning potential for this subset.

5 Code

[Github code link](#)